More Is Different, More Than Ever: Major Transitions From Biological Development to Planetary Intelligence

A Perspective

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Abstract

Biology is organized in layers within layers, where the components of one level form the substrate for the next higher level of organization. The collaborative principles behind such self-organizing systems – from symbiosis and symbiogenesis to collective and multiscale intelligence – may also be central to AI alignment. Departing from the picture of a genetic blueprint, modern interpretations view the genome as a generative model of organismal development: a compact, interactive library of evolutionarily encoded survival lessons of the past that is actively decoded and interpreted by cellular agents according to their multicellular developmental context. Large language models may be becoming something similar for cultural evolution: a compact, generative, interactable memory of collective human knowledge. In both cases, i.e., for cells or humans, this allows single agents to efficiently gain access to competencies they never acquired individually, and use them to coordinate their behavior within a more complex multicellular or multiagent Umwelt, respectively. In turn, this suggests that we may be approaching another major evolutionary "symbiomemetic" transition, driven by the co-evolution of biological and artificial intelligence. Rather than focusing exclusively on aligning or controlling a singular artificial general intelligence, we should therefore ask how humans, machines, institutions, ecosystems, and the biosphere can be integrated into a higher level of organization – a planetary superintelligence.

Keywords: Collective Phenomena, Evo-Devo, Generative Artificial Intelligence, Symbiogenesis, Co-Evolution, Co-Intelligence, Alignment

Running Title: LLMs as Generative Genome for Collective Human Knowledge

Traditionally, machine learning (ML) and artificial intelligence (AI) have been introduced into science as tools for dealing with complexity. Statistical ML techniques help us detect patterns, search rugged parameter spaces, and emulate complex systems. Recently, we have reached a tipping point with large language models (LLMs), and in particular with AI agents. Beyond helping us brainstorm hypotheses, swarms – or better, teams – of agents now write and verify code, and even conduct scientific research and write papers. While these developments are changing almost every field, this is not the deepest implication AI will have for science and society. For me, AI has not simply become a tool to accelerate my workflow. It has become a conceptual bridge to a question I had been circling for years: why is more different?[1]

As with many other interdisciplinary researchers, my scientific path can best be described as a random walk through domains – from statistical physics, to quantum optics and quantum information, soft and active matter, autonomous navigation of synthetic microorganisms and sailing boats, machine learning, biology, and artificial life. Ubiquitous in collective phenomena, seemingly independent parts give rise to coordinated behavior at the system level.

Like many physicists, I started with a reductionist reflex: chemistry is complicated physics, biology is a consequence of chemistry, and cognition is what happens when enough neurons are wired up in a brain. This is not wrong, but it deflects from a more fundamental domain-bridging

mechanism: at some point, more is different – not just more complicated, but different in kind. A cell is not just a bag of chemicals, a tissue is not just many cells, a forest not just many trees, and a mind not just many neurons – but the context matters. At some scale, encapsulated systems begin to actively maintain themselves against environmental perturbations. They physically separate their inside from the outside world, store information and respond to stressors to act in ways that increase their chances of persistence. Systems become agents.[2]

Even unicellular organisms consist of many different parts: molecular sensors, actuators, membranes, cytoskeletal structures, chemical networks, to name but a few. These parts must be coordinated just right for a cell to behave as a unit. Many such microorganisms move through viscous environments, where inertia is irrelevant and swimming requires coordinated body deformations. They achieve this without a central controller. No one is in charge, yet the microbe behaves "as if": it moves toward chemical sources, chases prey, avoids predators, etc.[3]

In simulations, we stripped this problem down to a deliberately minimal setting and treated such swimmers as decentralized collections of sensors and actuators. Each part senses only nearby information and acts locally. Under the right conditions, and using neuroevolution techniques, the whole swimmer learns, or evolves to swim collectively. The key point is not that the swimmer moves, but where the actual agent is located: not in a single component, but distributed across the dynamical relations of all parts.[4]

More thus becomes very different when parts collaborate so effortlessly that a higher level of organization, a top-tier agent acquires its own competencies. This is close to ideas from predictive processing and active inference:[5] persistent systems do not merely react to the external world, but act through constantly refined internal models of their relevant environment. This self-improving feedback loop of persisting through modeling, acting, and correcting sits at the center of how AI changed my view of biology, and eventually of intelligence and life itself.[6] While there may be no single "ghost in the machine",[7] in a sense, the shell itself incarnates the ghost.[8,9]

Biology is organized in layers within layers, where the components of one level form the substrate for the next. At each layer, components solve problems in their own domain. We usually measure intelligence through navigation in 3D space – because it is most familiar to us – but biology is full of problem-solving in much stranger spaces. Regenerative animals rebuild body parts, slime molds grow efficient transport networks, mycelium networks trade with tree roots, and cells can self-organize into novel lifeforms such as Xenobots or Anthrobots that are capable of healing neuronal tissue.[10,11] These examples are strange only if we expect intelligence to require a brain. Viewed differently, cells and cellular collectives are competent agents across metabolic, physiological, transcriptional, and anatomical spaces.[2]

Across scales, biology is working with unreliable hardware and software, from thermal noise in molecular systems or gene regulatory networks to behavioral interactions in organisms and group dynamics. Yet, multiscale biological systems are remarkably stable attractors in the morphogenetic landscape:[12] a tadpole transforms into a frog, a caterpillar into a butterfly, a kitten into a cat, and a puppy into a dog. Given the sheer complexity of a single organism, it is remarkable that life reproduces so reliably through a genomic bottleneck.

Importantly, the genome cannot encode every detail of an organism – every cell, vesicle, or hair. It encodes protein sequences and regulatory machinery that reconfigure how cells behave. Cells query the genome by transcribing genes and translating them into proteins, i.e., nanomachines that reconfigure cellular phenotypes under changing conditions. Modern interpretations go further and see the genome as compressed latent variables that instantiate organismal development:[13,14] an interactable library of survival lessons of the past, shaped by evolution and reinterpreted by cellular collectives during development. This points to a problem-solving layer between the genome and phenotypic form and function: life does not work with passive material, but with an agential cellular substrate that drives adaptation and evolvability.[2]

Biological self-organization should therefore not simply be described as self-assembling physics.

It is collective intelligence from the bottom up: plastic living matter learning how to persist in diverse forms by modeling the world across scales, while competing for resources, survival, and coherence. Strikingly, some of the most useful concepts for modeling such systems come from AI: autoencoders, diffusion models, and so-called neural cellular automata.[6,15,16]

Evolution, or natural selection, is often framed as a predominantely competitive process: good genes are selected from organisms with the best survival strategies.[17] The mentality of survival of the fittest has even found its way into the DNA of academia: publish or perish. But good survival traits are not necessarily competitive. Often, the exact opposite is the case. An overlooked aspect of evolution, and of life in general, is that it is fundamentally collaborative[18] – and this is where things become interesting: more can make a difference.

Symbiotic systems achieve remarkable things through specialization and division of labor: protein networks in cells, organs in bodies, neuronal dynamics in brains, species networks in ecosystems such as the wood-wide-web.[19] Strikingly, all of these examples resemble associative learning – something we typically discuss only in cognitive systems. The Hebbian idea of fire together wire together also fits evolutionary development: grow together evolve together.[20]

Sometimes, such symbiotic relations become permanent, and subsystems lose their ability to live or reproduce independently. Through symbiogenesis, living things are "becoming together", rather than simply living together, which brings forth completely novel survival strategies or life forms:[21–24] newly merged structures of organization that exploit independently evolved capabilities more effectively than the sum of the parts alone.

In several major evolutionary transitions,[25,26] biology leaped forward from a prebiotic molecular soup toward genomic encoding, from replicators toward chromosomes, from prokaryotes toward eukaryotes, from single cells toward multicellular organisms, from solitary organisms toward societies, and from animal communication toward language, culture, and technological civilization. Science often refers to this complexification process as emergence, but there seem to be regularities in how such novelty comes about in biology: it's about finding new levels of integration, where the dynamics at the macro-level are driven by the competencies of its parts. Something that was once an environment, partner, parasite, or neighbor becomes an integral part of something more: a new agent is formed.[27,28]

Evolution is tinkering.[29] It builds sophisticated problem-solving machines by repurposing and integrating previously acquired competencies under new contexts. Seen through this lens, evolution is literally a scaling mechanism of compositional problem solving, i.e., intelligence. Biology is therefore not simply layers within layers. It is models within models – agents all the way down, but also all the way up:[30] turtles both ways. Lower levels set the stage from the bottom up, but higher levels reshape top-down conditions under which lower levels operate: cells constrain molecules, tissues constrain cells, organisms constrain tissues, and society constrains us.

Our human mind can wrap itself around almost any process we can perceive or imagine, from attosecond physics, cosmology, to theory of mind, and *Alice in Wonderland* or *The Matrix*. We learn skills on demand, cooperate in large groups, divide labor, abstract fluidly, and achieve astonishing things collectively. Yet, although we like to imagine ourselves as the top of the pinacle of evolution, we remain part of the biosphere, held together by biological and increasingly technological layers.[31] Even our bodies are ecosystems of human cells and bacteria, in roughly comparable numbers. Moreover, only about 1-1.5% of our genome encodes proteins; large parts are transposable elements, and ancient viral remnants make up roughly 8%, at least.[32–34]

Contrary to popular belief, no single human is a general intelligence: we cannot learn everything within one lifetime, and some levels of complexity may simply be beyond our capabilities.[35] Many skills we take for granted – language, writing, math, agriculture, medicine, science, even pencils – can be acquired only through constant interaction with society. This casts cultural evolution in a different light: it is not separate from biology, but presumably the same evolutionary logic on a different substrate. It unfolds as a co-evolution of societies and human skill-acquisition,

progressing much faster than genetic evolution, i.e., through agents who coordinate, learn, and share what they know – and want to know.

Technology, especially computer science, pushes this further. Much of our collective knowledge is outsourced into books or servers, while supply chains and communication have been industrialized. Today, machines are no longer passive tools: they sense, compute, regulate, communicate, and act. Tools become infrastructure, which turns into dependency. Machines have become part of the environment we depend on, while technology depends on us. Yet again, this is symbiosis. Over evolutionary timescales, cells learned how to integrate into multicellular bodies, successively giving up independence to better integrate and benefit from the shared infrastructure of the multiscale organism. Something similar is happening to us. The transition into a cybernetically interconnected globalized society made us less independent individually, but much more capable collectively. The internet already forms something like a global nervous system for our technological civilization. But a nervous system is not automatically a mind. More generally, our current globalized society is powerful, but fragmented: we coordinate supply chains across continents, but fail to act collectively around system-level threats like climate change, war, inequality, and biodiversity loss. We are globally connected, but not yet globally coherent.[31]

AI changes the interface to this network, and with LLMs, we may be entering the next phase. Despite their shortcomings, LLMs are in many aspects already much more capable than any single human. They are not trained to mimic one persona, but on vast traces of human interaction: books, code, papers, chats, jokes, mistakes, images, etc. They do not model one mind but patterns in collective human knowledge and interaction. For the first time, we can poke the Zeitgeist and expect an answer back to the question: "what if?"

This is where the analogy to biology becomes more than a metaphor. Recall: the genome is a functional memory of evolutionary history that cells can query to reconfigure themselves under changing conditions. It does not encode every detail of the organism, but it gives cells and cellular collectives access to survival lessons they did not acquire individually. Agentic AI may be becoming something similar for cultural evolution: a compact, generative, interactable memory of collective human knowledge. It gives individual humans instant access to skills, concepts, and problem-solving strategies they did not personally learn, and maybe not even fully comprehend. For most of history, society was the reservoir from which humans drew skills. The internet made this reservoir globally accessible, and AI interactively integrates it into our everyday life: the collective knowledge of humankind at the tips of our fingers, asking "How can I help today?"[36–42]

For science, this is not merely an acceleration of our workflows. Of course, AI helps me write code, search literature, analyze complex data, model morphogenesis, evolve decentralized controllers, and build self-repairing synthetic systems. But more deeply, it forces us to stop treating intelligence as solely brain-centered, and life as purely carbon-based. The diverse intelligence movement[43,44] seeks common principles across these domains. It understands intelligence as a multiscale process: parts integrating into wholes without fully losing their individuality, which has strong implications for AI alignment[45].

The main challenge in AI may therefore not be to create a single artificial general intelligence (AGI); that idea is too narrow. The more interesting question is whether humans, machines, institutions, and the biosphere can form an ecosystem of striving biological and artificial agents that learns to coordinate at scale, while preserving the well-being of its constituents. Biology has repeatedly undergone such major evolutionary transitions, in which previously independent agents became parts of new higher-level individuals.[25,26] The growing integration of biological and artificial intelligence may be the beginning of another, "symbiomemetic", transition[27,28,31]— with LLMs, or more broadly agentic AI, in the role of a "generative cultural genome". What remains open is how this process can be coordinated, and toward what ends. If we play our cards right, we may not simply build an AGI; we may become part of a planetary superintelligence.

This brings us back to the original ambition of *Artificial Life* (ALife):[46] life, and intelligence, as they could be. The field is driven by its community's curiosity and pursuit of novelty, often without a particular engineering application in mind, but with a refreshing – good kind of – weirdness to it.[47] This open-ended, creative, and often rather playful exploration of "life-likeness" has long been an incubator for ideas that later matured across different disciplines, or spun-off entirely new ones. As biological and artificial agents increasingly become parts of the same systems, this scientific practice becomes paramount. Without having planned for it, ALife's open-ended tradition has spent decades developing the concepts, models, and experimental intuition needed to ask how new forms of agency coordinate across scales and substrates – which provides a powerful mindset, but also great responsibility, for shaping our collective future.

More is different. And individuals can make a real difference only if they learn how to collaborate at scale – by making the collective worth belonging to.

References

[1] Anderson PW. More Is Different: Broken symmetry and the nature of the hierarchical structure of science. Science. 1972;177(4047):393–396. Available from: http://dx.doi.org/10.1126/science.177.4047.393.

[2] Levin M. Darwin's agential materials: evolutionary implications of multiscale competency in developmental biology. Cellular and Molecular Life Sciences. 2023;80(6):142. Available from: https://doi.org/10.1007/s00018-023-04790-z.

[3] Purcell EM. Life at low Reynolds number. American Journal of Physics. 1977;45(1):3-11. Available from: https://doi.org/10.1119/1.10903.

[4] Hartl B, Levin M, Zöttl A. Neuroevolution of decentralized decision-making in N-bead swimmers leads to scalable and robust collective locomotion. Communications Physics. 2025;8(1). Available from: http://dx.doi.org/10.1038/s42005-025-02101-5.

[5] Friston K, FitzGerald T, Rigoli F, Schwartenbeck P, Pezzulo G. Active Inference: A Process Theory. Neural Computation. 2017;29(1):1–49. Available from: http://dx.doi.org/10.1162/NECO_a_00912.

[6] Hartl B, Pio-Lopez L, Fields C, Levin M. Remapping and navigation of an embedding space via error minimization: A fundamental organizational principle of cognition in natural and artificial systems. Physics of Life Reviews. 2026;58:336-59. Available from: https://www.sciencedirect.com/science/article/pii/S1571064526000527.

[7] Ryle G. The Concept of Mind. London: Hutchinson; 1949.

[8] Shirow M. Ghost in the Shell. Milwaukie, OR: Dark Horse Comics; 1995.

[9] Soma K, Bouteiller Y, Hamann H, Beltrame G. The hive mind is a single reinforcement learning agent. Swarm Intelligence. 2026;20(1):1–33. Available from: http://dx.doi.org/10.1007/s11721-026-00264-7.

[10] Kriegman S, Blackiston D, Levin M, Bongard J. A scalable pipeline for designing reconfigurable organisms. Proceedings of the National Academy of Sciences. 2020;117(4):1853-9. Available from: https://www.pnas.org/doi/abs/10.1073/pnas.1910837117.

[11] Gumuskaya G, Srivastava P, Cooper BG, Lesser H, Semegran B, Garnier S, et al. Motile Living Biobots Self-Construct from Adult Human Somatic Progenitor Seed Cells. Advanced Science. 2024;11(4):2303575. Available from: https://advanced.onlinelibrary.wiley.com/doi/abs/10.1002/advs.202303575.

[12] Waddington CH. Canalization of development and the inheritance of acquired characters. Nature. 1942;150(3811):563–565. Available from: http://dx.doi.org/10.1038/150563a0.

[13] Mitchell KJ, Cheney N. The Genomic Code: the genome instantiates a generative model of the organism. Trends in Genetics. 2025;41(6):462-79. Available from: https://www.sciencedirect.com/science/article/pii/S0168952525000083.

[14] Hartl B, Levin M. What does evolution make? Learning in living lineages and machines. Trends in Genetics. 2025;41(6):480-96. Available from: https://www.sciencedirect.com/science/article/pii/S0168952525000770.

[15] Mordvintsev A, Randazzo E, Niklasson E, Levin M. Growing Neural Cellular Automata. Distill. 2020;5(2). Available from: https://doi.org/10.23915/distill.00023.

[16] Hartl B, Levin M, Pio-Lopez L. Neural cellular automata: Applications to biology and beyond classical AI. Physics of Life Reviews. 2026;56:94-108. Available from: https://www.sciencedirect.com/science/article/pii/S1571064525001757.

[17] Darwin C. On the Origin of Species by Means of Natural Selection, or the Preservation of Favoured Races in the Struggle for Life. London: John Murray; 1859.

[18] Sheldrake M. Entangled life. London, England: Vintage; 2021.

[19] Simard SW, Perry DA, Jones MD, Myrold DD, Durall DM, Molina R. Net transfer of carbon between ectomycorrhizal tree species in the field. Nature. 1997;388(6642):579–582. Available from: http://dx.doi.org/10.1038/41557.

[20] Watson RA, Szathmáry E. How Can Evolution Learn? Trends in Ecology & Evolution. 2016;31(2):147-57. Available from: https://doi.org/10.1016/j.tree.2015.11.009.

[21] Barricelli NA. Esempi numerici di processi di evoluzione. Methodos. 1954;6(21-22):45-68.

[22] Barricelli NA. Symbiogenetic Evolution Processes Realized by Artificial Methods. Methodos. 1957;9(35–36):143-82.

[23] Kozo-Polyansky BM. Symbiogenesis: A New Principle of Evolution. Fet V, Margulis L, editors. Harvard University Press; 2010. Available from: http://dx.doi.org/10.2307/j.ctv22jnrvc.

[24] Margulis L. Symbiogenesis. A new principle of evolution rediscovery of Boris Mikhaylovich Kozo-Polyansky (1890–1957). Paleontological Journal. 2010;44(12):1525–1539. Available from: http://dx.doi.org/10.1134/S0031030110120087.

[25] Szathmáry E, Smith JM. The major evolutionary transitions. Nature. 1995;374(6519):227–232. Available from: http://dx.doi.org/10.1038/374227a0.

[26] Szathmáry E. Toward major evolutionary transitions theory 2.0. Proceedings of the National Academy of Sciences. 2015;112(33):10104–10111. Available from: http://dx.doi.org/10.1073/pnas.1421398112.

[27] Ashford J, Cvjetko M, Löffler R, Sakallioglu B, Valerio A, Tataryn M, et al.. Evolving Symbiosis, from Barricelli's Legacy to Collective Intelligence: a simulated and conceptual approach; 2026. Available from: https://arxiv.org/abs/2603.08463.

[28] Ashford J, Sakallioglu B, Tataryn M, Valerio A, Hartl B, Pio-Lopez L, et al. Symbiogenesis: from Barricelli's Legacy to Collective Intelligence. In: Proceedings of the Genetic and Evolutionary Computation Conference Companion. GECCO '26 Companion. New York, NY, USA: Association for Computing Machinery; 2026. p. 1228–1236. Available from: https://doi.org/10.1145/3795101.3814707.

[29] Jacob F. Evolution and Tinkering. Science. 1977;196(4295):1161-6. Available from: https://www.science.org/doi/abs/10.1126/science.860134.

[30] Levin M, Watson R. Machines all the way up and cognition all the way down: Updating the machine metaphor in biology. Seminars in Cell & Developmental Biology. 2026;177-178:103668. Available from: https://www.sciencedirect.com/science/article/pii/S1084952126000029.

[31] Agüera y Arcas B. What is Intelligence?: Lessons from AI about Evolution, Computing, and Minds. MIT Press; 2025.

[32] Sender R, Fuchs S, Milo R. Revised Estimates for the Number of Human and Bacteria Cells in the Body. PLOS Biology. 2016;14(8):e1002533. Available from: http://dx.doi.org/10.1371/journal.pbio.1002533.

[33] Dunham I, Kundaje A, Aldred SF, Collins PJ, Davis CA, Doyle F, et al. An integrated encyclopedia of DNA elements in the human genome. Nature. 2012;489(7414):57-74. Available from: https://doi.org/10.1038/nature11247.

[34] Griffiths DJ. Endogenous retroviruses in the human genome sequence. Genome Biology. 2001;2(6):reviews1017.1. Available from: https://doi.org/10.1186/gb-2001-2-6-reviews1017.

[35] Kahneman D. Thinking, Fast and Slow. New York, NY: Farrar, Straus & Giroux; 2011.

[36] OpenAI. Introducing ChatGPT; 2022. https://openai.com/index/chatgpt/.

[37] Anthropic. How Claude Code works;. Accessed 3 November 2026. Claude Code documentation. Available from: https://code.claude.com/docs/en/how-claude-code-works.

[38] Bonamy N, Choi D. Codex as a platform: build on the open agent harness; 2026. Accessed 3 November 2026. OpenAI Developers. Available from: https://developers.openai.com/blog/codex-as-a-platform.

[39] Zechner M, et al.. Pi: A minimal, extensible agent harness;. Accessed 3 November 2026. GitHub software repository. Available from: https://github.com/earendil-works/pi.

[40] Steinberger P. Introducing OpenClaw; 2026. Accessed 3 November 2026. OpenClaw Blog. Available from: https://openclaw.ai/blog/introducing-openclaw.

[41] Nous Research. Hermes Agent;. Accessed 3 November 2026. GitHub software repository. Available from: https://github.com/NousResearch/hermes-agent.

[42] Lange RT, Grillotti L, Yamada Y, Charakorn R, Ha D. The Agentic Scientist: Orchestrating Coding Agents for Automated Scientific (Co-)Discovery; 2026. Available from: https://arxiv.org/abs/2610.18933.

[43] Levin M. Technological Approach to Mind Everywhere: An Experimentally-Grounded Framework for Understanding Diverse Bodies and Minds. Frontiers in Systems Neuroscience. 2022 Mar;16. Available from: http://dx.doi.org/10.3389/fnsys.2022.768201.

[44] Levin M. Artificial Intelligences: A Bridge Toward Diverse Intelligence and Humanity's Future. Advanced Intelligent Systems. 2025 Apr;7(10). Available from: http://dx.doi.org/10.1002/aisy.202401034.

[45] Lyons B, Pio-Lopez L, Levin M. Alignment Is to a Virtual Governor: A Theory of Coordination in Diverse Intelligence. Preprints. 2026. Available from: https://doi.org/10.20944/preprints202607.0220.v1.

[46] Langton CG, editor. Artificial life. Complex Adaptive Systems. Cambridge, MA: Bradford Books; 1995.

[47] Stanley KO, Lehman J. Why greatness cannot be planned: The myth of the objective. Springer; 2015.